NANDINI AGRAWAL

B.E. Electronics & Telecommunication Engineering

+91-9130694591 — nandiniagraw259@gmail.com — linkedin.com/in/nandini-agrawal08 — Pune, India

Professional Summary

Electronics Telecommunication Engineering student with hands-on experience in embedded systems, IoT, and full-stack software development. Proficient in Java, SQL and C/C++ with a strong foundation in DBMS and hardware-software integration. Passionate about building reliable software solutions and leveraging technology to solve real-world problems.

Education

International Institute of Information Technology (I^{2}IT), Pune — *2023 – Present*
B.E. Electronics & Telecommunication Engineering — **CGPA: 8.72/10**

Little Flowers School & Junior College, Gondia — *2022 – 2023*
HSC - Class 12 — **66.17%**

Gondia Public School, Gondia — *2020 – 2021*
CBSE - Class 10 — **77.67%**

Technical Skills

Programming Languages	C, C++, Java, Python, SQL
Core Computer Science	Object-Oriented Programming (OOP), DBMS, Computer Networks, Operating Systems (Basic), Real-Time Systems
Embedded Systems & IoT	Ultrasonic Sensors, Servo Motors, DC Motors, Relay Modules, LCD Display, Solenoid Valve, Electromagnet, Relay Control,Schematic Design & Interpretation
Tools & Technologies	Git, GitHub, MySQL, IntelliJ IDEA, Visual Studio Code, Arduino IDE, Proteus, Wokwi, Multisim
Technical Concepts	Hardware Testing & Debugging, Technical Documentation, Report Preparation, ERP Documentation

Work Experience

Intern – ERP Software Development — *Sep 2025 – Nov 2025*
512 Army Base Workshop, Kirkee (Government of India)

- Prepared functional and technical documentation for an ERP system supporting the Indian Army (EME).
- Documented business workflows, software requirements, and process flows in collaboration with cross-functional teams.
- Assisted in requirement analysis and maintained project documentation throughout the development lifecycle.

Projects

Automatic Electronic Component Retrieval Robot — *Jan 2026 – May 2026*

- Designed and developed an autonomous robot for electronic component detection and retrieval using **ESP32, ultrasonic sensors, DC motors, servo motor, and an electromagnet**.
- Integrated **hardware components, implemented control logic** and performed **system testing and debugging** to ensure reliable operation.

Smart Washroom Automation System — *Jan 2025 – May 2025*

- Developed an Arduino-based smart washroom automation system using **ultrasonic sensors, relay-controlled flushing, solenoid valve, and LCD display** to optimize water usage.
- Performed hardware integration, testing, and debugging to validate system functionality.
- Co-authored an **IEEE conference paper** based on the project.

Smart Home Automation System *Jan 2024 – May 2024*

- Designed and developed an IoT-based home automation system using **ESP8266 NodeMCU, relay modules, and Blynk IoT** for remote monitoring and control of electrical appliances.
- Integrated Wi-Fi communication, performed **hardware testing and debugging**, and ensured stable system performance.

Leadership & Extracurricular

- **Head – Young Indians (Yi) Committee:** Led the planning and execution of student workshops, networking events, and club meetings, collaborating with cross-functional teams to ensure successful event management.
- **Lady Representative – ISETS (I^2IT Society of Electronics & Telecommunication Students):** Coordinated communication between students and faculty and assisted in organizing departmental events.
- **Social Media Manager – ATAGTR Pune:** Managed digital content and collaborated with the team to increase website traffic and grow social media followers by [number].